

PRAKTIKUM SISTEM OPERASI

**MODUL IX
SYSCALL XINU**



Disusun Oleh :

Alya Rabani / 2311104076

SE-07-02

Dosen Pengampu :

Yudha Islami Sulistya, S.Kom., M.Cs.

PROGRAM STUDI S1 SOFTWARE ENGINEERING

FAKULTAS INFORMATIKA

TELKOM UNIVERSITY PURWOKERTO

2025

JURNAL

1. Buat syscall baru seperti yang ditunjukkan pada modul syscall poin 9.5! (sertakan Screenshot kode dan hasil run)

- a. Membuat syscall baru

```
// Nama: Alya Rabani
// NIM: 2311104076

#include <xinu.h>

/*-----
 * chname - Dummy syscall
 *-----
 */
syscall chname(pid32 pid, char *newname) {
    intmask mask;

    mask = disable();

    /* Validasi PID */
    if (isbadpid(pid)) {
        restore(mask);
        return SYSERR;
    }

    /* Validasi nama */
    if (newname == NULL) {
        restore(mask);
        return SYSERR;
    }

    /* Ubah nama proses */
    strncpy(proctab[pid].pname, newname, PNMLEN-1);
    proctab[pid].pname[PNMLEN-1] = '\0';

    restore(mask);
    return OK;
}
```

- b. Menambahkan prototypes.h

```
extern status ascdate(uint32, char *);

/* in file bufinit.c */
extern status bufinit(void);

/* in file chprio.c */
extern syscall chprio(pid32, pri16);

/* in file chname.c */
extern syscall chname(pid32, char *);

/* in file clkupdate.S */
extern uint32 clkcount(void);

/* in file clkhandler.c */
extern interrupt clkhandler(void);

/* in file clkinit.c */
extern void clkinit(void);

/* in file clkdisp.S */
extern void clkdisp(void);

/* in file close.c */
extern syscall close(did32);

/* in file control.c */
extern syscall control(did32, int32, int32, int32);

/* in file create.c */
extern pid32 create(void *, uint32, pri16, char *, uint32, ...);

/* in file ctxsw.S */
extern void ctxsw(void *, void *, void *);
```

- c. xsh_uptime

```

/* xsh_uptime.c - xsh_uptime */

#include <xinu.h>
#include <stdio.h>
#include <string.h>

/*-----
 * xsh_uptime - shell to print the time the system has been up
 *-----
 */
// Alya Rabani
// 2311104076
shellcmd xsh_uptime(int nargs, char *args[])
{
    uint32  days, hrs, mins, secs; /* Days, hours, minutes, and */
                                     /* seconds since system boot */
    uint32  secperday = 86400;      /* Seconds in a day */
    uint32  secperhr  = 3600;       /* Seconds in an hour */
    uint32  secpermin  = 60;        /* Seconds in a minute */

    /* Nama proses baru */
    char prosesBaru[] = "ProcBaru";

    /* For argument '--help', emit help about the 'uptime' command */
    if (nargs == 2 && strcmp(args[1], "--help", 7) == 0) {
        printf("Use: %s\n", args[0]);
        printf("Description:\n");
        printf("\tDisplays time since the system booted\n");
        printf("Options:\n");
        printf("\t--help\t display this help and exit\n");
        return 0;
    }

    ----- days, hours, mins, secs, -----

    /* Subtract number of hours */
    hrs  = secs/secperhr;
    secs -= hrs*secperhr;

    /* Subtract number of minutes */
    mins = secs/secpermin;
    secs -= mins*secpermin;

    printf("Xinu has been up ");
    if (days > 0) {
        printf(" %d day(s) ", days);
    }

    if (hrs > 0) {
        printf(" %d hour(s) ", hrs);
    }

    if (mins > 0) {
        printf(" %d minute(s) ", mins);
    }

    if (secs > 0) {
        printf(" %d second(s) ", secs);
    }
    printf("\n");

    /* Syscall baru*/
    kprintf("Mengubah nama proses ID 5 Menjadi %s...\n", prosesBaru);
    chname(5, prosesBaru);

    return 0;
}

```

d. Output

```
Activities Terminal Fri 1 May, 21:03
xinu@xinu-develop-end: ~/xinu/compile

File Edit View Search Terminal Help

116598 bytes of Xinu code.
[0x00100000 to 0x0011C775]
148392 bytes of data.
[0x0011FF20 to 0x001442C7]

Obtained IP address 10.0.3.15 (0x0a00030f)

-----ALYA RABANI-----

XINU
-----2311104076-----

Welcome to Xinu!

xsh $ uptime
Xinu has been up 1 second(s)
Mengubah nama proses ID 5 Menjadi ProcBaru...
xsh $
CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Offline | ttyS0
```

2. Perbaiki syscall chprio (xinu/system/chprio.c) dengan memperhatikan validasi input

- Pastikan id adalah angka dari 0 – NPROC (ukuran maks banyaknya proses)
- Pastikan prioritas adalah bilangan yang positif

Compile dan jalankan Xinu dengan syscall yang telah diperbaiki

- make
- clean

```
// Nama: ALYA RABANI
// NIM: 2311104076

#include <xinu.h>

/*-----
 * chprio - Change the scheduling priority of a process
 *-----*/
syscall chprio(pid32 pid, pri16 newprio)
{
    intmask mask;
    struct procent *prptr;
    pri16 oldprio;

    mask = disable();

    if (pid < 0 || pid >= NPROC) {
        restore(mask);
        return SYSERR;
    }
    if (newprio <= 0) {
        restore(mask);
        return SYSERR;
    }
    prptr = &proctab[pid];
    oldprio = prptr->prprio;
    prptr->prprio = newprio;

    restore(mask);
    return oldprio;
}
```

```

Welcome to Xinu!

xsh $ ps
Pid Name          State Prio Ppid Stack Base Stack Ptr Stack Size Msg Content
-----
0 prnull          ready  0    0 0x005FDFFC 0x00FFFEF0 8192 1 3
1 ipout           wait  500   0 0x005FBFFC 0x005FBF30 8192 0 0
2 netin           wait  500   0 0x005F9FFC 0x005F9ED0 8192 0 0
4 Main process    recv  20    3 0x005E7FFC 0x005E7F34 65536 0 0
5 shell           recv  50    4 0x005F7FFC 0x005F79AC 8192 0 0
6 ps              curr  20    5 0x005F5FFC 0x005F5FC4 8192 0 0

xsh $ uptime
Xinu has been up 47 second(s)
Mengubah nama proses ID 5 Menjadi ProcBaru...
xsh $ ps
Pid Name          State Prio Ppid Stack Base Stack Ptr Stack Size Msg Content
-----
0 prnull          ready  0    0 0x005FDFFC 0x00FFFEF0 8192 1 3
1 ipout           wait  500   0 0x005FBFFC 0x005FBF30 8192 0 0
2 netin           wait  500   0 0x005F9FFC 0x005F9ED0 8192 0 0
4 Main process    recv  20    3 0x005E7FFC 0x005E7F34 65536 0 0
5 ProcBaru        curr  50    4 0x005F7FFC 0x005F79AC 8192 0 7
8 ps              curr  20    5 0x005F5FFC 0x005F5FC4 8192 0 0

xsh $
CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Online 0:0 | ttyS0

```

3. Lakukan hal-hal berikut ini

Edit xsh_uptime.c

- Tambahkan kode berikut

```

/* Subtract number of hours */
hrs = secs/secperhr;
secs -= hrs*secperhr;

/* Subtract number of minutes */
mins = secs/secpermin;
secs -= mins*secpermin;

printf("Xinu has been up ");
if (days > 0) {
    printf(" %d day(s) ", days);
}

if (hrs > 0) {
    printf(" %d hour(s) ", hrs);
}

if (mins > 0) {
    printf(" %d minute(s) ", mins);
}

if (secs > 0) {
    printf(" %d second(s) ", secs);
}
printf("\n");

/* Syscall baru */
kprintf("Mengubah prioritas proses ID 5 menjadi 33...\n");
chprio(5, 33);

return 0;
}

```

Compile source code tersebut dengan perintah

- make clean
- make

Jalankan perintah ps

- xsh \$ ps
- perhatikan prioritas proses dengan id = 5

Jalankan uptime

- xsh \$ uptime

- Perhatikan hasil perintah tersebut

Jalankan ps

- xsh \$ ps
- perhatikan prioritas proses dengan id = 5 seharusnya sudah berubah

```

Activities  Terminal  Fri 1 May, 21:34
xinu@xinu-develop-end: ~/xinu/compile

Welcome to Xinu!

xsh $ ps
Pid Name          State Prio Ppid Stack Base Stack Ptr  Stack Size Msg  Content
-----
0 prnull          ready  0    0 0x005FDFFC 0x00FFFE0 8192 1 3
1 ipout           wait  500  0 0x005FBFFC 0x005FBF30 8192 0 0
2 netin           wait  500  0 0x005F9FFC 0x005F9ED0 8192 0 0
4 Main process    recv  20   3 0x005E7FFC 0x005E7F34 65536 0 0
5 shell           recv  50   4 0x005F7FFC 0x005F79AC 8192 0 0
6 ps             curr  20   5 0x005F5FFC 0x005F5FC4 8192 0 0

xsh $ uptime
Xinu has been up 8 second(s)
Mengubah prioritas proses ID 5 menjadi 33...
xsh $ ps
Pid Name          State Prio Ppid Stack Base Stack Ptr  Stack Size Msg  Content
-----
0 prnull          ready  0    0 0x005FDFFC 0x00FFFE0 8192 1 3
1 ipout           wait  500  0 0x005FBFFC 0x005FBF30 8192 0 0
2 netin           wait  500  0 0x005F9FFC 0x005F9ED0 8192 0 0
4 Main process    recv  20   3 0x005E7FFC 0x005E7F34 65536 0 0
5 shell           recv  33   4 0x005F7FFC 0x005F79AC 8192 0 7
8 ps             curr  20   5 0x005F5FFC 0x005F5FC4 8192 0 0

xsh $
CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Online 0:1 | ttyS0

```

Testing chprio syscall yang telah diubah

- Testing prioritas tidak boleh < 0: Ubah “chprio(5,33)” menjadi “chprio(5,-3)” pada xsh_uptime.c

```

/* Subtract number of hours */
hrs = secs/secperhr;
secs -= hrs*secperhr;

/* Subtract number of minutes */
mins = secs/secpermin;
secs -= mins*secpermin;

printf("Xinu has been up ");
if (days > 0) {
    printf(" %d day(s) ", days);
}

if (hrs > 0) {
    printf(" %d hour(s) ", hrs);
}

if (mins > 0) {
    printf(" %d minute(s) ", mins);
}

if (secs > 0) {
    printf(" %d second(s) ", secs);
}
printf("\n");

/* Syscall baru*/
kprintf("Mengubah prioritas proses ID 5 menjadi 33...\n");
chprio(5, -3);

return 0;
}

```

```

Activities Terminal Fri 1 May, 21:46
xinu@xinu-develop-end: ~/xinu/compile
File Edit View Search Terminal Help

Welcome to Xinu!

xsh $ ps
Pid Name      State Prio Ppid Stack Base Stack Ptr  Stack Size Msg  Content
-----
0 prnull      ready  0    0 0x005FDFFC 0x00FFFE0 8192 1 3
1 ipout       wait  500  0 0x005FBFFC 0x005FBF30 8192 0 0
2 netin       wait  500  0 0x005F9FFC 0x005F9ED0 8192 0 0
4 Main process recv  20   3 0x005E7FFC 0x005E7F34 65536 0 0
5 shell       recv  50   4 0x005F7FFC 0x005F79AC 8192 0 0
6 ps         curr  20   5 0x005F5FFC 0x005F5FC4 8192 0 0

xsh $ uptime
Xinu has been up 5 second(s)
Mengubah prioritas proses ID 5 menjadi 33...
xsh $ ps
Pid Name      State Prio Ppid Stack Base Stack Ptr  Stack Size Msg  Content
-----
0 prnull      ready  0    0 0x005FDFFC 0x00FFFE9C 8192 1 3
1 ipout       wait  500  0 0x005FBFFC 0x005FBF30 8192 0 0
2 netin       wait  500  0 0x005F9FFC 0x005F9ED0 8192 0 0
4 Main process recv  20   3 0x005E7FFC 0x005E7F34 65536 0 0
5 shell       recv  50   4 0x005F7FFC 0x005F79AC 8192 0 7
6 ps         curr  20   5 0x005F5FFC 0x005F5FC4 8192 0 0

xsh $
CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Online 0:0 | ttyS0

```

- Testing id adalah valid: Ubah “chprio(5,33)” menjadi “chprio(3000,3)”

```

/* Subtract number of hours */
hrs = secs/secperhr;
secs -= hrs*secperhr;

/* Subtract number of minutes */
mins = secs/secpermin;
secs -= mins*secpermin;

printf("Xinu has been up ");
if (days > 0) {
    printf(" %d day(s) ", days);
}

if (hrs > 0) {
    printf(" %d hour(s) ", hrs);
}

if (mins > 0) {
    printf(" %d minute(s) ", mins);
}

if (secs > 0) {
    printf(" %d second(s) ", secs);
}
printf("\n");

/* Syscall baru*/
kprintf("Mengubah prioritas proses ID 5 menjadi 33...\n");
chprio(3000, 3);

return 0;
}

```

```
Activities Terminal Fri 1 May, 21:54
xinu@xinu-develop-end: ~/xinu/compile
File Edit View Search Terminal Help

Welcome to Xinu!

xsh $ ps
Pid Name      State Prio Ppid Stack Base Stack Ptr  Stack Size Msg  Content
-----
0 prnull      ready  0    0 0x005FDFFC 0x00FFFE0 8192 1 3
1 ipout       wait   500  0 0x005FBFFC 0x005FBF30 8192 0 0
2 netin       wait   500  0 0x005F9FFC 0x005F9ED0 8192 0 0
4 Main process recv   20  3 0x005E7FFC 0x005E7F34 65536 0 0
5 shell       recv   50  4 0x005F7FFC 0x005F79AC 8192 0 0
6 ps         curr   20  5 0x005F5FFC 0x005F5FC4 8192 0 0

xsh $ uptime
Xinu has been up 7 second(s)
Mengubah prioritas proses ID 5 menjadi 33...
xsh $ ps
Pid Name      State Prio Ppid Stack Base Stack Ptr  Stack Size Msg  Content
-----
0 prnull      ready  0    0 0x005FDFFC 0x00FFFE0 8192 1 3
1 ipout       wait   500  0 0x005FBFFC 0x005FBF30 8192 0 0
2 netin       wait   500  0 0x005F9FFC 0x005F9ED0 8192 0 0
4 Main process recv   20  3 0x005E7FFC 0x005E7F34 65536 0 0
5 shell       recv   50  4 0x005F7FFC 0x005F79AC 8192 0 7
8 ps         curr   20  5 0x005F5FFC 0x005F5FC4 8192 0 0

xsh $
CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Online 0:0 | ttyS0
```

- Hasil dua testing di atas adalah prioritas tidak berubah karena salah argument (dibuktikan dengan menggunakan perintah ps)